

Vectors and Coordinates #2

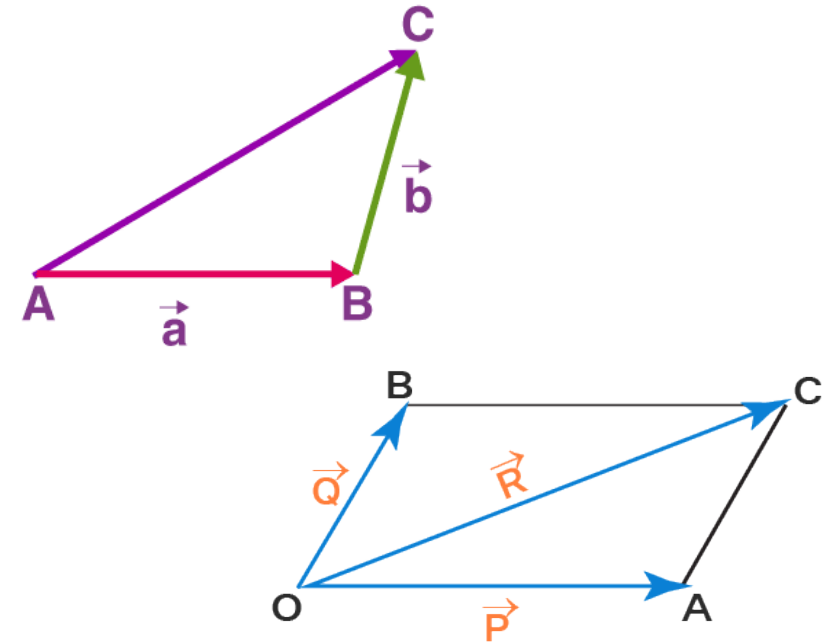


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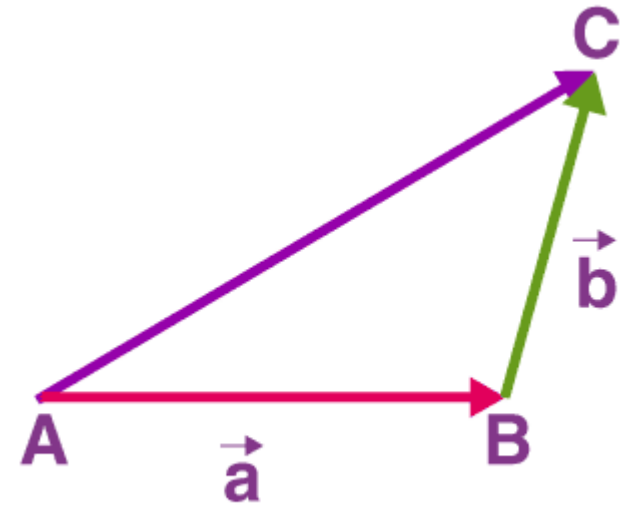
Adding Vectors

- There are several methods to add two vectors together
- There are two main graphical methods:
 - Tip-to-Tail Method
 - Parallelogram Method
- And one main mathematic method:
 - Component addition
- **Note graphical methods only work when drawing to scale**



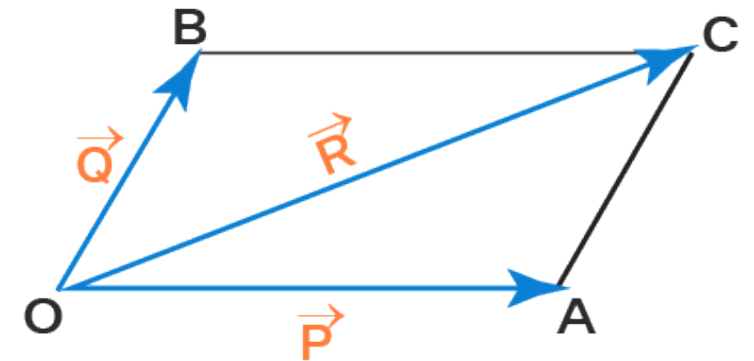
Tip to Tail Method

- In tip to tail method you draw one vector on a graph
- Then draw the next vector extending from the tip of the original
- We can repeat this for however many vectors you want to add
- Then we can measure the final line and work out the angle using a ruler and a protractor



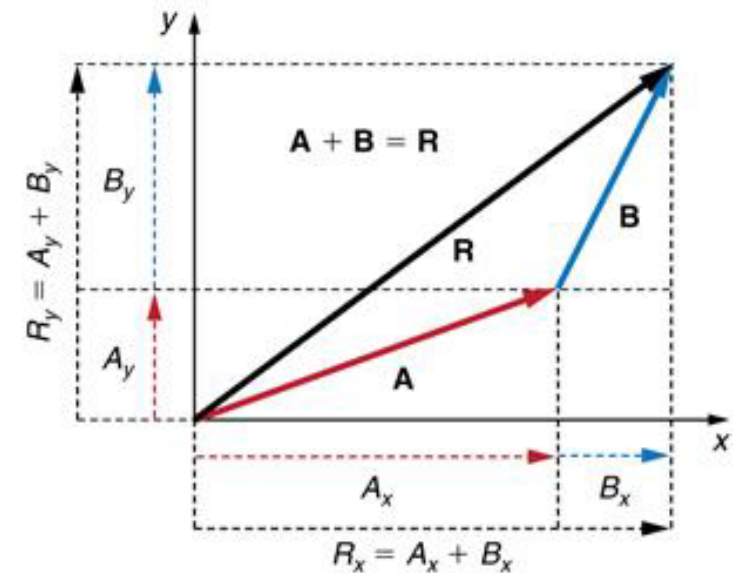
Parallelogram Method

- In the parallelogram method we draw out the vectors we are adding starting at the 0 point
- We then take these two lines to draw a parallelogram
- We then work out the distance from the 0 point to the far end of that parallelogram
- We can also work out the angle for this line for the resultant vector



Component Addition

- For component addition we split both vectors into X and Y
- We then add both Xs and both Ys to give us a final resultant cartesian vector
- We can then do cartesian to polar conversion to find a resultant vector



Component addition example

- We can look at the graph on this page, we know that A is (6,1)
- We can also determine that B is (1,6)
- So, if we add our x components: $6+1 = 7$
- Then if we add our y components: $1+6 = 7$
- Then we get the final vector (7,7)

